The Holographic Alaya–Field Framework: World, Interface and Subject

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General Abstract

How do physical interactions form a stable world, the interfaces through which it can be reached, and systems whose histories organize their own participation in it? The Holographic Alaya–Field Framework studies these three questions together. Its central proposal is that usable structure depends on relations of access that can form, persist, be shared and be revised. A channel's resolution, a protected algebra, a finite task capability and a spacetime geometry are different objects. The theory connects them through specified interactions, records and reconstruction maps.

The world line begins with a distinction that an algebra alone can miss. In HAFF A, channels with the same output algebra can resolve different information. A classification of qubit channels with equal pure-input entropies, finite probe tests, and one common reconstruction rule separate those differences. In a three-qubit example, physical reductions of the same initial preparation give nonisometric finite correlation geometries. Their couplings can change the reduced states and three-body coherence; the comparison holds the initial source and reconstruction rule fixed, not the final physical state. It does not derive every spacetime metric [1].

HAFF B and E supply dynamics and instruments to this account. Couplings can select stable blocks that carry both classical labels and protected quantum content. Their finite lifetime matters as much as the exact fixed algebra: a task's risk can vary continuously even when that algebra changes discontinuously. A probe's allowed effects, the records it actually acquires, and the effects that persist need not coincide. Environmental fragments can disclose block labels without disclosing the quantum content within them. The models give concrete conditions for repeatable and public reading [2, 5].

A system can then use records to maintain its ability to intervene. HAFF C defines this capacity through an actual body, a resource budget and a causal process. Histories that leave the same current observation can warrant different actions; learning a diagnostic relation can alter the body's later capability. This gives agency a physical history and a maintained boundary [3]. HAFF F follows the same distinction through time. Acquiring new records, retaining old correlations and renewing degraded memories have different entropy accounts. Correlated copies can retain a label with zero marginal purity. Noisy cyclic correction removes a specified joint entropy and, for a stated thermal exhaust, has a finite-rate heat price. These results explain an operational record direction and the

conditions for continued activity while keeping the physical clock and initial resource boundary explicit [6].

HAFF D connects this account to gravity along two routes: conditional modular/holographic reconstruction and the actual gravitational response of access-changing matter. Local field constructions match full spatial quantum sources, including pressure and clock energy. A finite-bandwidth mean-action model closes a source–geometry–sensor–force return on a prepared local patch. The full-mode quartet supplies a common nonlinear bulk solution for the retained state, physical clock, material director and constrained JT source under one mean action. General gravitational generation remains a strong mechanism hypothesis: selected structure, source, response and future permissions should share compatible dynamics. Joining the full-mode bulk to physical endpoints, the complete gate schedule and sustained renewal remains a construction task [4].

The physical interface has its own six-paper line, Q-RAIF. A derives Lorentzian and Clifford readout under local propagation and single-shell conditions, and tests when species share a cone. B studies the actual operations and resources needed to protect, control and read content. C follows quantum records into material motion and interacting field sources under a positive autonomous Hamiltonian; its finite receiver-update family retains the old receiver, references and spent correlations. Mass–Energy connects spectral mass to finite physical setting and readout. Null Cluster compares null and non-null access in one closed source task with matched resources. No View identifies source components that a given readout cannot recover, and supplies additional physical access where it closes that blind spot. Positive control cost, information metrics and Lorentzian spacetime are not identified by their shared mathematical form [8].

T-DOME asks how these interactions become a system's own organization. I establishes when a retained record is worth its preparation and upkeep in a specified action task. II shows how finite history selects a self-frame: a retained relation between the task and the body's channels. An aligned axis alone can leave prediction useless; outcome-bearing memory supplies the signed relation needed to use it. III studies how a system distinguishes world change from its own miscalibration and uses actual references, body history and value to choose inquiry and maintenance. The memory–self–introspection route thus has task-specific necessity results, constructive mechanisms and failure regimes. It proposes a continuing, revisable subject without demanding perfect self-knowledge [9].

Foundation tests existence, sharing and return. I constructs stable hulls from a source, noise and readable seed, with conditions for geometric inheritance. II separates shared carrier, state, agreement and correctness. III gives a finite semiclassical mean-action scalar/JT return: an acquired record protects original entangled content and changes its apparatus's source; the response enters an X-basis reading of the old probe into the spent cell. Its first-order geometric signal arises after record acquisition. The compared runs share the same initial mass and geometric constraints; old references and the cursor remain in the joint state. Separate quantum-material constructions renew receivers and install any specified finite acquisition/decision/replacement schedule on one physical cursor, or refresh a full-mode body and its mixed receiver/purifier with delayed incoming correlations and complete matrix stress. These retain different prepared banks, frames and endpoints. They support finite return and renewal, not a single sustained gravitational/thermal realization; that remains the stronger conjecture [10, 8].

Consciousness carries an additional explanatory responsibility. I relates geometry to modular structure in exact supplied net models and develops the significance of a selected perspective. II asks how physically prepared sources and readable modular data constrain

higher-dimensional geometric laws. III studies finite self-readout and the continuing observer's first-person organization. The latter requires an experience bridge with its own reasons, competing explanations and interventions. Functional calibration, report differences and an account of experience answer different questions; a useful self-model does not settle all three [11].

Imprint and Brain Anchor make the history in this account physically consequential. Imprint distinguishes carrying a trace, reading it through a reference, combining it with other traces, and using it to change the next inquiry or act. Its scale models quantify what survives filtering and when retained history helps after the present loses predictive sufficiency. Relational integration models show when information becomes useful joint content; the chain connects records to subsequent maintenance and action [12]. Brain Anchor supplies a body mechanism: joint sensorimotor history learns membership and relative orientation, changes routing and can install protection for content already held. Learning and migration consume that content's lifetime. Longer training can therefore improve identification while worsening survival. These are finite physical formation mechanisms and biological intervention hypotheses, not neural evidence for a quantum code [13].

HAFF G returns to the question of internal execution. Program-controlled couplings can select protected relations without every program becoming a central label. Records can select their own protection and change subsequent matter sources. Exact and approximate programming results price the available readouts; finite noise limits how long history-selected sources differ. The supplied executor remains a separate origin problem. The broader proposal is that adaptive bodies acquire operational permissions through retained relations, then use those permissions to revise themselves [7].

Together these results develop a theory in which a history is part of what makes later access and action possible. They provide exact constructions, conditional laws and stronger formation hypotheses at different scales. The finite quantum body, the classical self-frame, the algebraic net and the gravitational response have compatible questions but do not yet form one universal implementation. Primitive dynamics, general geometry, unique measurement outcomes and experience retain their independent explanatory burdens. The postscript draws out a consequence that survives these differences: a trace can persist after active participation ends, while a continuing subject can live with incomplete self-knowledge [14].

Keywords: physical access, quantum channels, stable structure, records, self-frames, maintained agency, shared worlds, physical feedback

References

All numeric record identifiers below denote DOIs with the prefix `10.5281/zenodo.`; they identify the series records, not public uploads of these working revisions.

[1] S. Liu, *Physical Access and Correlation Geometry: The Holographic Alaya–Field Framework*, (2026), series record DOI: 10.5281/zenodo.18361706.

[2] S. Liu, *Coupling, Records, and Stable Relations: Formation, Protected Content and the Lifetime of a Geometry*, (2026), series record DOI: 10.5281/zenodo.18367060.

[3] S. Liu, *Causation, Agency, and Existence: Historical Control and Maintained Intervention*, (2026), series record DOI: 10.5281/zenodo.18374805.

[4] S. Liu, *Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return*, (2026), series record DOI: 10.5281/zenodo.18388881.

[5] S. Liu, *Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout*, (2026), series record DOI: 10.5281/zenodo.18400065.

[6] S. Liu, *Records, Dissipation and Intervention: Conditions for a Temporal Direction*, (2026), series record DOI: 10.5281/zenodo.18400425.

[7] S. Liu, *Internal Selection and Revisable Organization: How History Changes Permissions and Physical Sources*, (2026), series record DOI: 10.5281/zenodo.18402907.

[8] S. Liu, *Q-RAIF*, six October 2026 working papers: A (18525876), B (18525890), C (18528934), Mass–Energy (20280951), Null Cluster (20283057), and No View (20284454).

[9] S. Liu, *T-DOME*, October 2026 working trilogy: I, *Memory as a Usable Resource* (DOI: 10.5281/zenodo.18574342); II, *Reference Frames Selected by History* (18579703); III, *Keeping a Self-Model Calibrated* (18591771).

[10] S. Liu, *Foundation*, October 2026 working trilogy: I, *From Source to Manifestation* (DOI: 10.5281/zenodo.19627786); II, *Shared Manifestation* (19641372); III, *The Perfuming Loop* (19642189).

[11] S. Liu, *Consciousness*, October 2026 working trilogy: I, *Geometry Relative to a Cut* (DOI: 10.5281/zenodo.19607387); II, *Higher-Dimensional Consistency of Geometry and Modular Flow* (19607543); III, *Finite Self-Models and the Explanatory Gap* (19607724).

[12] S. Liu, *Imprint*, four October 2026 working papers: *The World as Imprint* (DOI: 10.5281/zenodo.20122718); *Renormalization and Imprint Filtering* (20151451); *Stable Hulls as Imprint Integration* (20151508); *The Imprint Chain* (20151594).

[13] S. Liu, *The Brain as a Plastic Observable-Algebra Network: Learning Shared Access from Sensorimotor History*, (2026), series record DOI: 10.5281/zenodo.20044329.

[14] S. Liu, *Postscript: Persistence, Participation and an Unfinished Self*, (2026), series record DOI: 10.5281/zenodo.18407367.